

Backfill: Trading the Suppliers Behind Drug Shortages

1. Summary

Backfill investigates whether drug shortages create investment opportunities in the listed suppliers that remain able to supply the affected product. The disrupted manufacturer identifies the shock & the proposed investment basket contains potential beneficiaries.

We reconstruct historical FDA shortage records, identify available suppliers, connect them to their listed parents, and evaluate long supplier positions hedged with SPY. Our current candidate waits 20 trading sessions after the first eligible close and holds for the next five sessions.

In the June 2014–September 2024 development sample, the candidate produced a net Sharpe ratio of **0.535** across eight positions and seven events. On an average a separate October 2024 Hurricane Helene illustration produced a **+1.06% net hedged return** for an equal-weight FMS/ICUI beneficiary basket.

2. Economic Hypothesis and Research Development

When a manufacturer cannot supply a drug, demand may shift to other manufacturers of the relevant product. Our hypothesis is that this redistribution can improve the beneficiaries' commercial outlook before investors fully incorporate it into their share prices.

Identifying the potential beneficiary requires more than reading a shortage headline. It requires connecting the affected formulation to supplier availability, manufacturing ownership and the correct listed company. This creates a plausible information-processing friction: the commercial implications are more difficult to interpret than the initial disruption.

Procurement adjustments may also delay the effect. Hospitals, distributors and purchasing organizations need to assess alternatives and redirect orders. We investigate whether beneficiary returns concentrate in a delayed, short window. The precise 20-session delay is a research-selected rule, rather than an independently established industry wide procurement lag.

Our initial specification expected a more immediate and persistent supplier response, tested through a 60-session holding period. That backtest did not support the initial specification. We therefore refined the approach toward stricter availability evidence and a delayed five-session holding window.

Our distinctive contribution is turning temporary regulatory records into an auditable supplier-to-stock signal. FDA research documents cases in which existing or new suppliers increased production after shortages, providing an economic basis for investigating this channel. [FDA shortage report](#)

The prediction concerns **positive aggregate net hedged returns for qualified beneficiaries**, rather than a guaranteed gain for every supplier. The hypothesis fails if comparable controls perform similarly, independent events do not reproduce the effect, or realistic execution costs absorb it.

3. Data and Universe

The historical development period covers **June 2014–September 2024**. We trade US-listed supplier-parent equities and use SPY as the market benchmark and hedge. Return accounting uses daily adjusted prices.

The reference backtest uses Webull data with recorded Yahoo Finance fallbacks for TEVA. A Webull-only comparison is reported separately to show the effect of vendor coverage.

Historical FDA data required reconstruction. FDA states that resolved shortages remain on its webpage for **six months**, while discontinued products remain for **one year**. The live database therefore cannot reproduce a complete historical supplier state. We use archived shortage lists and formulation detail pages, preserving capture timestamps and the underlying supplier evidence. FDA also notes that presentation availability is manufacturer-reported and can change daily. [FDA database FAQ](#)

We map manufacturers to listed parents using dated ownership and listing information. Missing predecessor prices, ambiguous mappings and unusable captures lead to exclusions. Their allocated capital remains cash, avoiding artificial increases in the weights of the remaining stocks.

4. Methodology

Event identification: An event occurs when an ingredient first appears as currently in shortage in the archived records after at least 180 days of absence. Supplier matching retains the formulation identifier.

Supplier qualification: Within the fixed 30-day archive matching window, we select the latest preceding supplier capture, or the earliest subsequent capture when no preceding capture exists. Every recorded presentation for the selected parent on that page must be available, unallocated and nonempty. This screen verifies the recorded availability state. It does not establish spare manufacturing capacity or exact commercial substitutability.

Information and execution timing: The information date is the later of the shortage observation and supplier capture. Subsequent evidence is never backdated to the initial notice.

The first eligible execution is the next US trading-session close. The candidate enters **20 additional trading sessions after that close** and exits **five session intervals after entry**. Supplier status is not reverified during the waiting period.

Position construction: Each event receives **5% of portfolio NAV**, divided equally among selected US suppliers before price exclusions. Each position retains its own expiry. The disrupted manufacturer receives no allocation in that event's beneficiary basket.

Market hedge. We estimate each supplier's beta using 250 paired stock/SPY returns ending before entry. That beta remains fixed throughout the position.

For a position without risk adjustments:

$$R_{\text{net}} = R_{\text{supplier}} - \beta R_{\text{SPY}} - C_{\text{trading}} - C_{\text{carry}}$$

Costs are expressed relative to initial stock capital. Portfolio returns reflect actual allocations and idle cash.

Cost assumptions: We charge 10 basis points per side for stocks, 2 basis points per side for SPY, and 50 basis points annually for hedge carry. The stress scenario doubles both trading costs and carry. Requests, exclusions, trades, lot cash flows and daily NAV are retained for reproduction.

5. Results

Historical Development Sample

All figures below include modeled transaction costs and hedge carry. Annualized portfolio metrics include inactive periods and idle cash.

Metric	Baseline costs	Doubled costs
Annualized return	0.105%	0.097%
Annualized volatility	0.198%	0.197%
Sharpe ratio	0.535	0.495
Maximum drawdown	-0.219%	-0.220%
Annualized turnover, both legs	12.87%	12.87%
Mean net hedged lot return	2.84%	2.59%
Positions / economic events	8 / 7	8 / 7

Six of eight positions were profitable under baseline costs. The portfolio was active on only **35 of 2,601 sessions**, explaining why individual event returns are substantially larger than annualized portfolio returns.

The Webull-only comparison retained four positions across three events and produced a net Sharpe of **0.609**. This represents a smaller coverage sample.

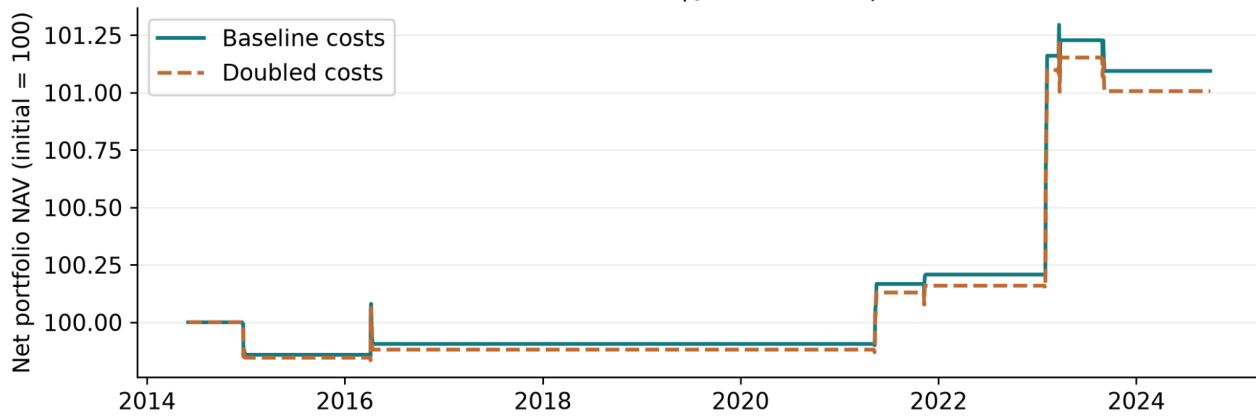


Figure 1: historical net portfolio equity, including idle cash, at baseline and doubled costs.

Later-Period Evidence and the Helene Illustration

The strict October 2024–October 2026 archive audit produced no eligible trades. Archive gaps and missing usable supplier evidence prevented a qualifying sample, so strict later-period performance metrics are unavailable.

We separately examined Hurricane Helene as an illustration of the economic mechanism. **Baxter identified the disruption and received zero portfolio weight.** On October 10, 2024, Fresenius Medical Care stated that it was working to increase IV and peritoneal-dialysis production following the disruption. [Company statement](#)

FMS and ICUI formed the related-supplier comparison. Applying the previously evaluated delay and holding lengths gave entry on **November 11** and exit on **November 18, 2024**.

Beneficiary	Net SPY-hedged return	Doubled costs
FMS	4.13%	3.90%
ICUI	-2.01%	-2.28%
Equal-weight beneficiary basket	1.06%	0.81%

The positive aggregate result is consistent with the directional hypothesis despite one supplier losing money. It includes the contribution of the short-SPY hedge. At a 5% total event allocation, the basket contributes approximately **+0.053% to portfolio NAV**.

6. Risk Management

New entries are constrained by:

- **8%** single-name allocation.
- **30%** allocation to one market and **50%** to one sector.
- **150%** gross exposure and **25%** absolute net exposure.

Following a 10% portfolio drawdown, exposure is halved from the next session and restored after 20 sessions. Positions otherwise exit on their scheduled dates; shortage resolution does not trigger an early exit.

The SPY hedge reduces estimated broad-market exposure but leaves company, healthcare-sector, liquidity and overnight-gap risks. Historical maximum marked gross exposure was **10.76%**. Low observed drawdown under sparse exposure does not establish resilience across stressed regimes.

7. Liquidity and Capacity

We estimate trading participation using 60 preceding sessions of dollar volume and examine both stock and hedge legs at entry and exit.

A **1%-of-average-daily-volume screen** implies a limiting historical portfolio size of approximately **\$0.46 million**. This is a volume-based sizing bound, rather than demonstrated profitable capacity.

Indicative spreads sometimes exceed the fixed cost assumptions. Before implementation, execution costs need calibration using dated bid–ask spreads and market-impact estimates. Doubled-cost results measure sensitivity; they do not verify actual fills.

8. Limitations and Next Steps

The evidence comes from a small, research-selected sample. Archived availability may be stale by entry, and formulation matching does not prove sufficient capacity or exact substitutability. Control comparisons have not established a statistically reliable advantage, and the revised candidate has no untouched out-of-sample confirmation.

The next step is a prospective test on newly observed events with rules frozen before outcomes. Daily supplier capture, availability checks at entry, improved product matching and calibrated execution costs would determine whether the observed relationship becomes a repeatable trading opportunity.

— — — — Thanks, Team BackFill.